Gravity Energy Storage Systems

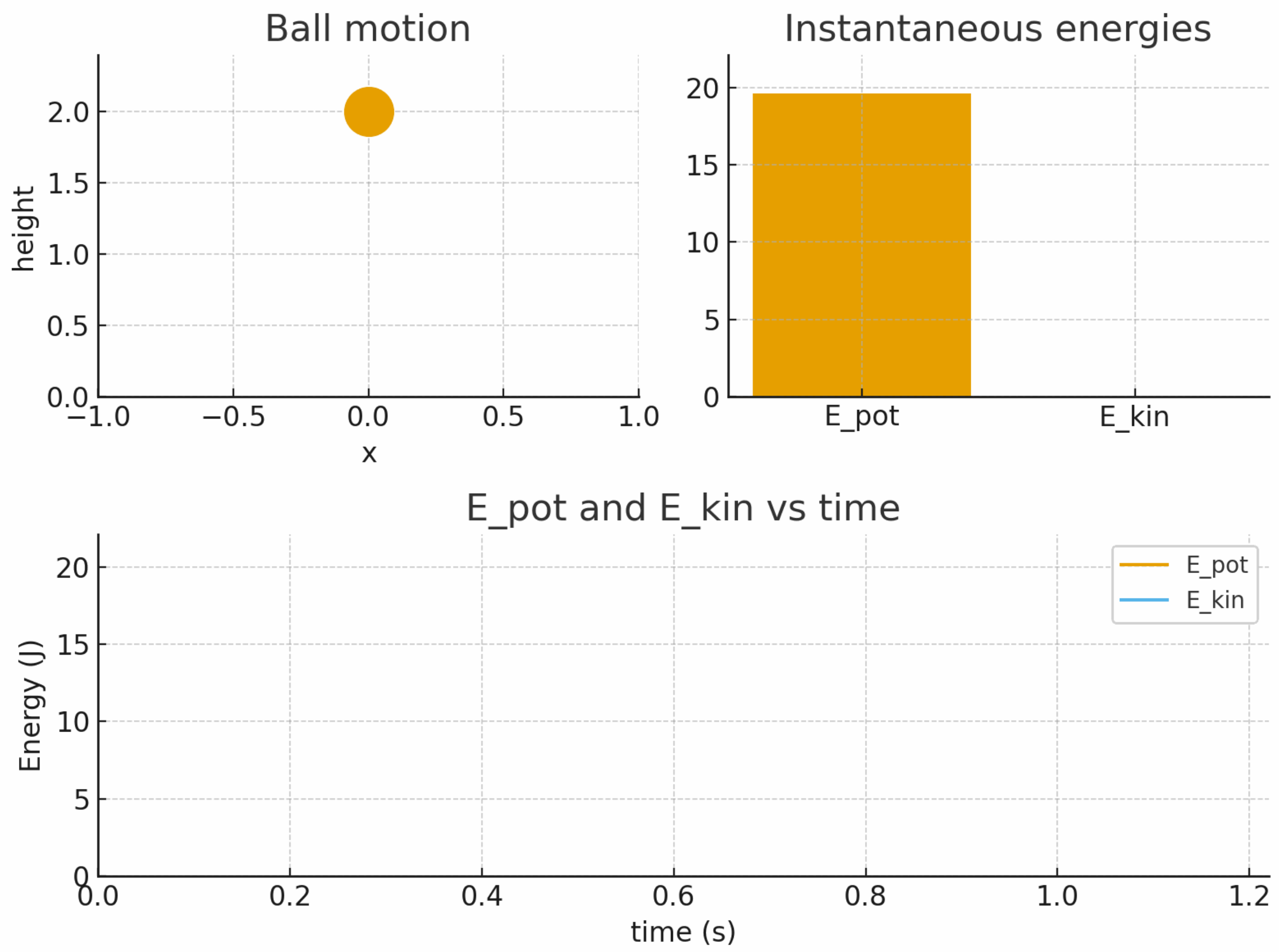
EENG Presentation – HS25

Matteo Frongillo & Michael Moos

Source: Energy Vault®

Gravity-based Reversible Energy Conversion

- Potential <=> Kinetic energy
- $E_{pot} = m \cdot g \cdot \Delta z$
- $E_{kin} = \dfrac{m \cdot v^2}{2}$
- High round-trip efficiency
- No degradation in time

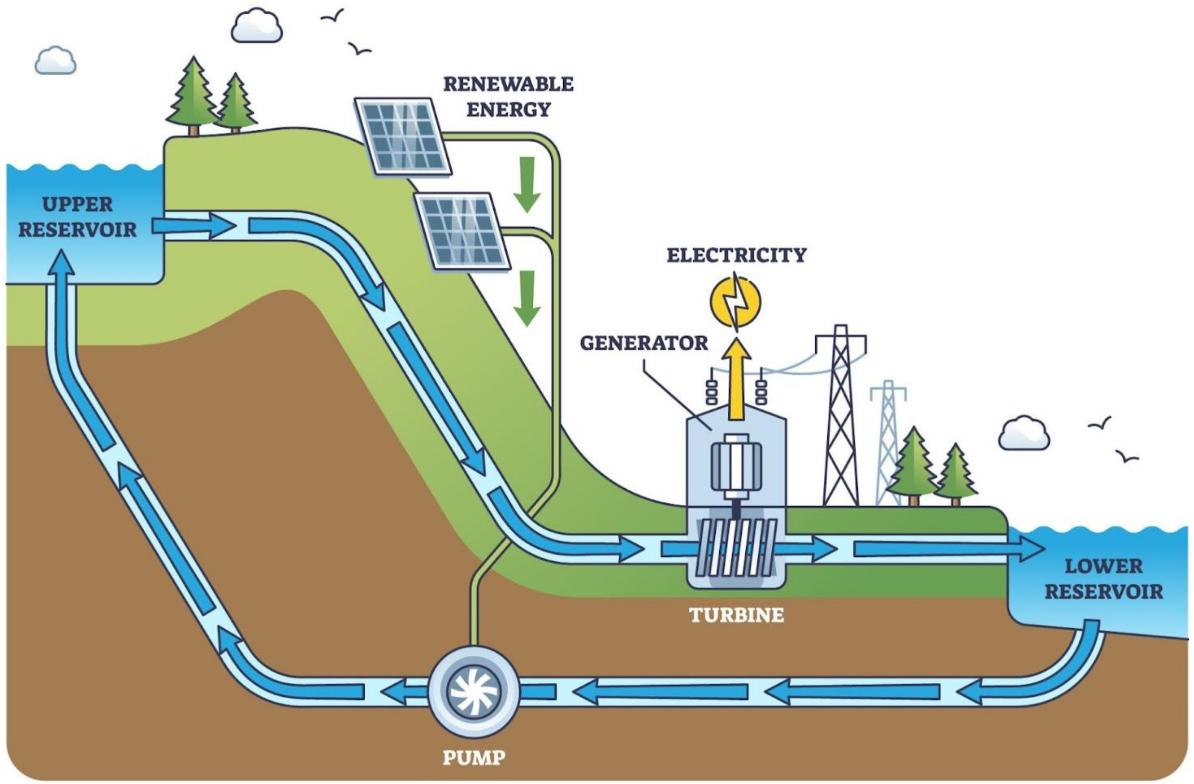


Gravity Energy Storage in Switzerland

Type of storage:
Pumped-hydro storage

Type of generation:
Hydroelectricity

Location:
Alps, mountains, rural areas



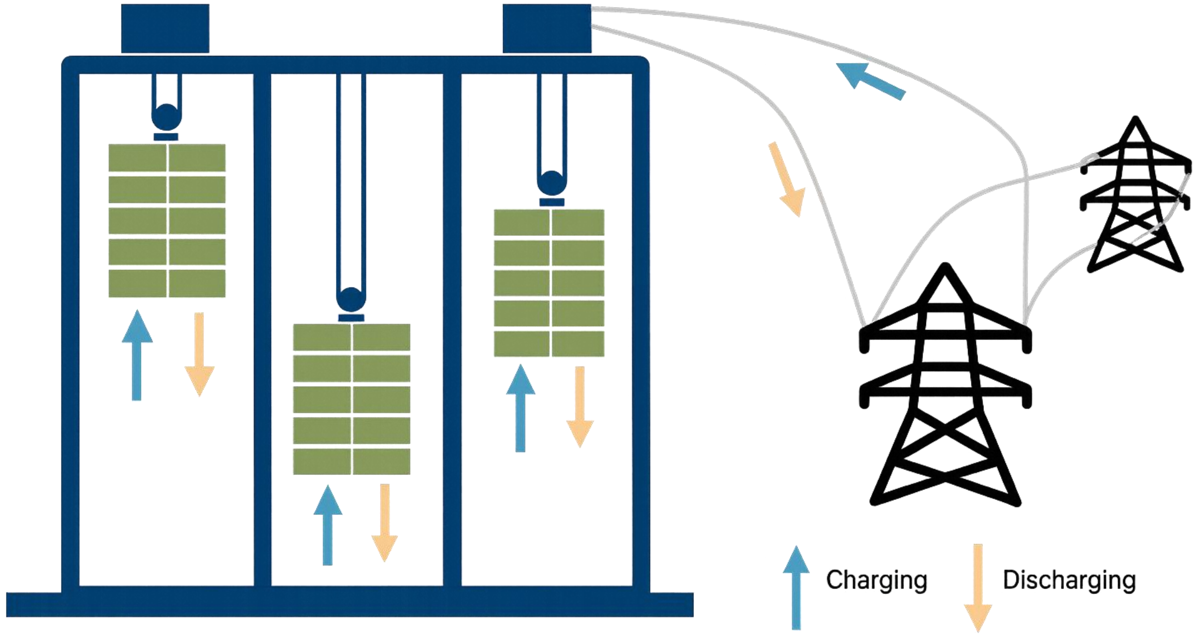
Source: Johnson Ye, ResearchGate

Gravity Energy Storage variant

Type of storage:
Gravity potential storage

Type of generation:
Electrical energy

Location:
Urban areas, outskirts



Source: Sateesh Chelimila, Aranca

GESS Variant

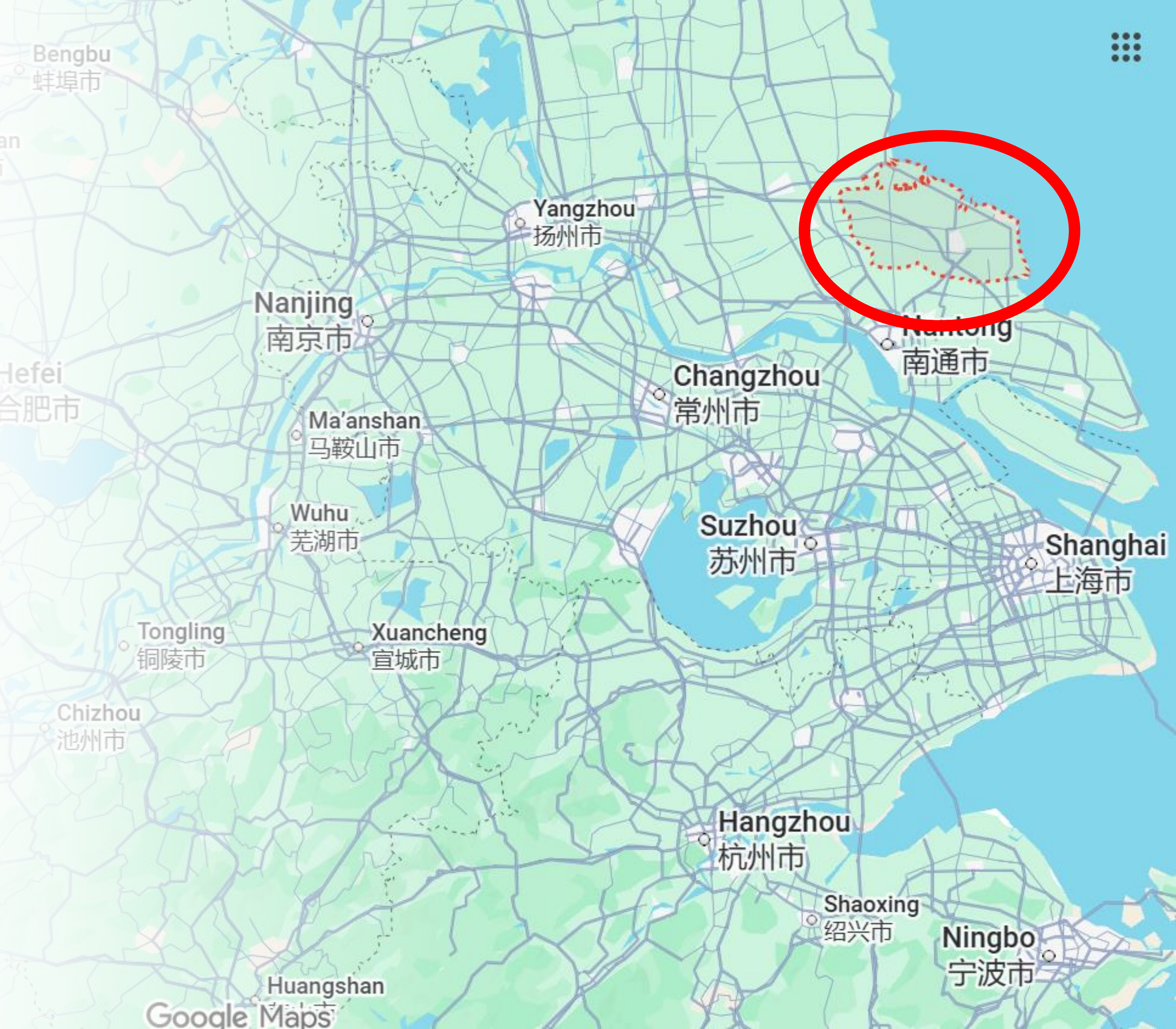
Source: Prototype 1, Energy Vault®

Industry Example: GESS Rudong, China

Source: Energy Vault®

GESS Rudong, Jiangsu, China

- First non-pumped GESS of commercial scale worldwide
- Capacity: 100 MWh
- Peak power output: 25 MWh over 4 hours
- Technology: EVx™ Gravity Energy Storage System



Source: Google Maps

How it is built

- Modular EVx tower design
- Automated lifting and lowering system
- Controlled by VaultOS™ (Energy Vault's software platform)
- Uses locally manufactured blocks (cheaper & sustainable)
- Round-trip efficiency: >80%
- Design lifetime: 35+ years

Source: Energy Vault®

Rudong Daily Operation

Excess Solar and Windpower

EVx uses excess energy to lift the blocks

Electricity demand spikes

EVx lowers the blocks and feeds power back into the grid

Source: Energy Vault®

Importance For China

MASSIVE RENEWABLE EXPANSION → HUGE FLUCTUATIONS

EXPENSIVE LITHIUM BATTERIES, AGE OVER TIME, AND RELY ON CRITICAL MINERALS

GRAVITY SYSTEMS CAN BE BUILT ANYWHERE

LOWER LONG-TERM MAINTENANCE COSTS

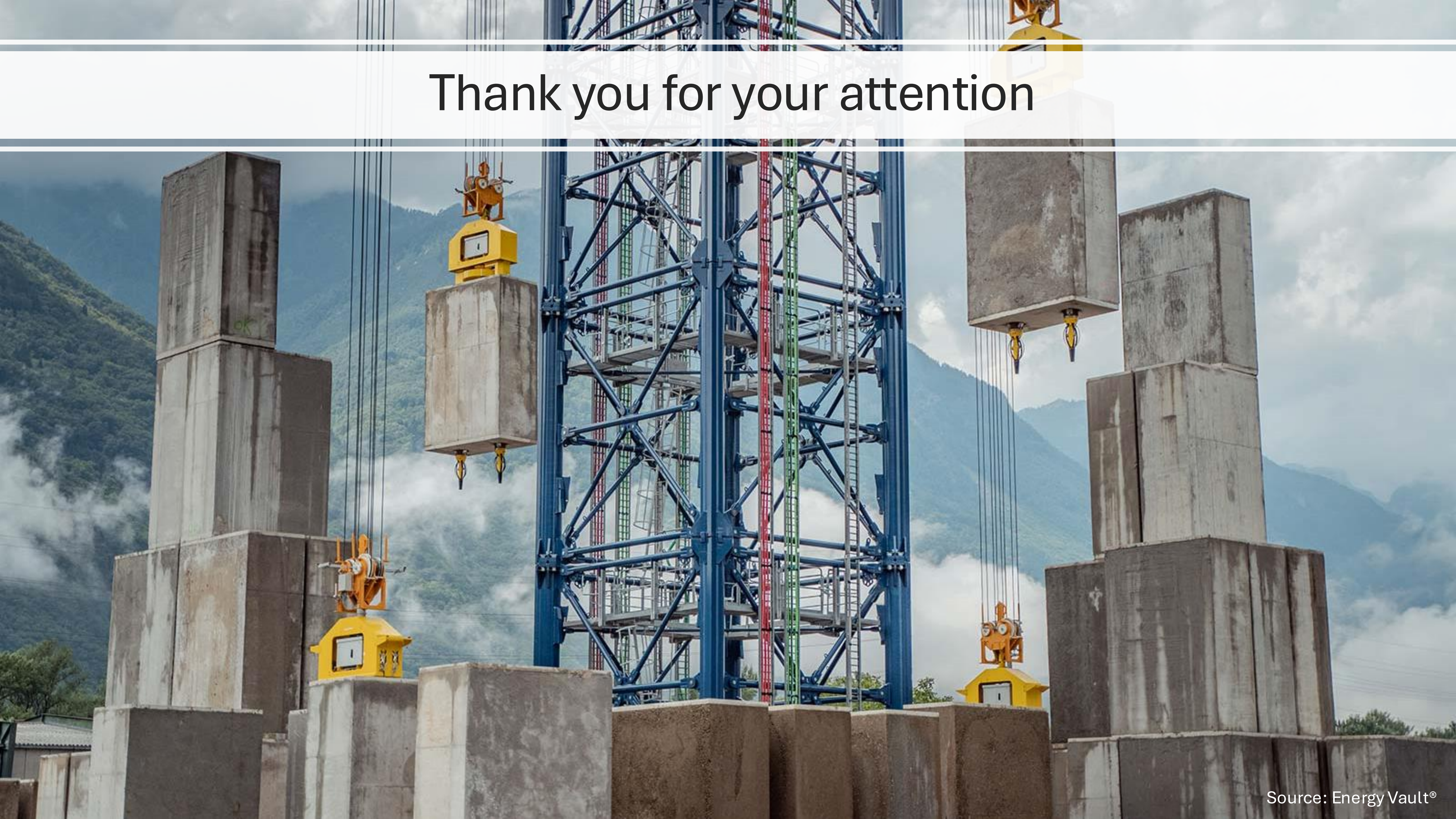
FITS PERFECTLY INTO CHINA’S ‘30-60 CARBON NEUTRALITY PLAN’

Industrial Importance

- First commercial proof-of-concept
- If Rudong performs well → scalable globally
- Competes with long-duration storage (flow batteries, compressed air)
- Advantages: long life, stable performance, sustainable materials, degradation free

Source: Energy Vault®

Thank you for your attention



Source: Energy Vault®